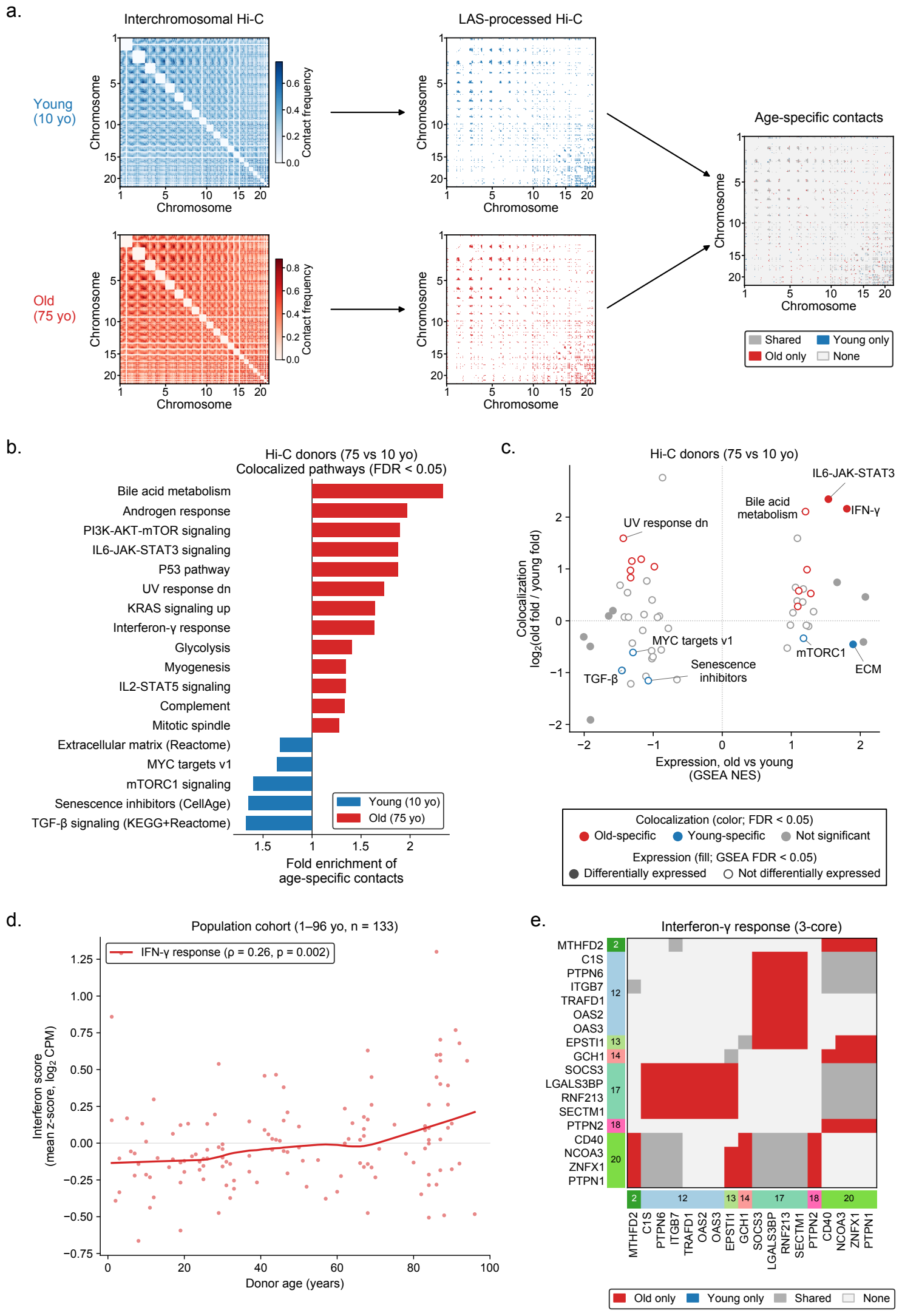

a.
Interchromosomal Hi-C
LAS-processed Hi-C
Young (10 yo)
Old (75 yo)
Chromosome
Contact frequency
Age-specific contacts
Shared
Young only
Old only
None
b.
Hi-C donors (75 vs 10 yo)
Colocalized pathways (FDR < 0.05)
Bile acid metabolism
Androgen response
PI3K-AKT-mTOR signaling
IL6-JAK-STAT3 signaling
P53 pathway
UV response dn
KRAS signaling up
Interferon-γ response
Glycolysis
Myogenesis
IL2-STAT5 signaling
Complement
Mitotic spindle
Extracellular matrix (Reactome)
MYC targets v1
mTORC1 signaling
Senescence inhibitors (CellAge)
TGF-β signaling (KEGG+Reactome)
Young (10 yo)
Old (75 yo)
Fold enrichment of age-specific contacts
c.
Hi-C donors (75 vs 10 yo)
IL6-JAK-STAT3
Bile acid metabolism
IFN-γ
UV response dn
MYC targets v1
mTORC1
ECM
Senescence inhibitors
TGF-β
Colocalization log2(old fold / young fold)
Expression, old vs young (GSEA NES)
Colocalization (color; FDR < 0.05)
Old-specific
Young-specific
Not significant
Expression (fill; GSEA FDR < 0.05)
Differentially expressed
Not differentially expressed
d.
Population cohort (1–96 yo, n = 133)
IFN-γ response (ρ = 0.26, p = 0.002)
Interferon score (mean z-score, log2 CPM)
Donor age (years)
e.
Interferon-γ response (3-core)
MTHFD2
C1S
PTPN6
ITGB7
TRAFD1
OAS2
OAS3
EPSTI1
GCH1
SOCS3
LGALS3BP
RNF213
SECTM1
PTPN2
CD40
NCOA3
ZNFX1
PTPN1
Old only
Young only
Shared
None